A Leakage-Free Model Comparison for Orange Juice Brand Choice

Lianjie (Marc) Shi · Machine Learning and Artificial Intelligence, Peking University · October 2026
Code: github.com/Marclianjie/data-science-portfolio/tree/main/oj-brand-choice

---

1. Problem

The task is to predict whether a shopper buys Citrus Hill (CH) or Minute Maid (MM) from 1,070 purchases (the *OJ* data distributed with ISLP; Stine, Foster & Waterman, 1998), using 17 predictors that describe prices, discounts, promotions, store, week and the customer's loyalty to Citrus Hill. The aim was not only a good classifier but an evaluation protocol that cannot leak information from the test set into any modeling decision.

2. Protocol

- **Split.** A stratified 80/20 split (856 training and 214 test purchases; 61% CH in both). The test set stayed untouched until the final model was fixed.
- **Pipelines.** All preprocessing (one-hot encoding of 3 categorical predictors, scaling of 14 numeric ones) lives inside scikit-learn `Pipeline`s, so it is refit within each fold; every model uses the same five stratified folds.
- **Models.** Grid-searched logistic regression, k-nearest neighbors, a decision tree and gradient boosting, compared against a majority-class baseline, with bias–variance curves for the number of neighbors and the tree depth.
- **Redundancy.** Nine of the 17 predictors duplicate information carried by others (store recodings and derived price columns). A compact eight-predictor set was compared with the expanded set.
- **Final checks.** The selected model was evaluated once on the test set, then stress-tested with a temporal split (trained on weeks 227–269, evaluated on weeks 270–278) and interpreted with coefficients and permutation importance.

3. Results

Model (tuned)	CV accuracy	ROC-AUC	Macro-F1
Logistic regression (C = 0.1)	82.2% ± 3.6	0.888	0.810
Gradient boosting (100 trees, learning rate 0.1)	81.3% ± 4.6	0.880	0.802
Decision tree (depth 4, min. leaf 20)	80.4% ± 4.7	0.859	0.792
k-nearest neighbors (k = 15)	78.7% ± 3.7	0.839	0.772
Majority baseline (always CH)	61.0%	0.500	0.379

Table 1. Five-fold cross-validated performance on the training set (mean ± SD of accuracy across folds).

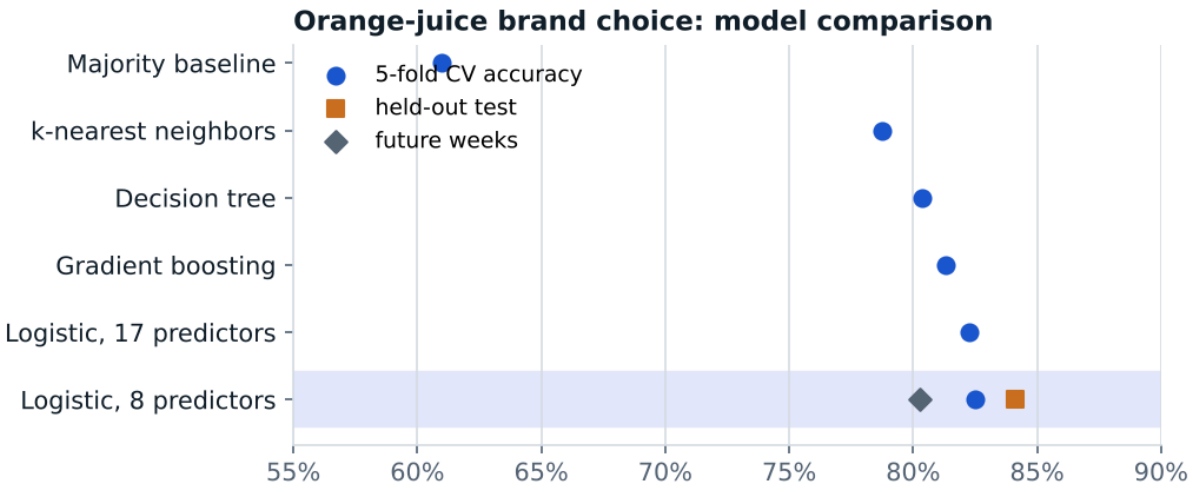


Figure 1. Cross-validated accuracy by model, with the final model's held-out test and future-week accuracy.

Compact features lose nothing. With the eight-predictor set, logistic regression reached 82.5% cross-validated accuracy (82.2% with all 17) and gradient boosting 81.3% (81.3%). The compact logistic regression (C = 0.1) was therefore fixed as the final model before the test set was opened.

Test and future performance. On the test set the final model reached 84.1% accuracy, 0.92 ROC-AUC and 0.83 macro-F1, against a 61% majority baseline (119 of 131 CH and 61 of 83 MM purchases classified correctly). Trained on early weeks and evaluated on later ones, it reached 80.3% accuracy and 0.89 ROC-AUC. Loyalty to Citrus Hill dominates the predictions, followed by the price gap between the two brands.

4. Limitations

- Fold-to-fold variation (3–5 points) is larger than the gaps between the top three models, so the ranking among them is not decisive; logistic regression was chosen because it was best on average and also the simplest.
- The drop from 84.1% on the random test split to 80.3% on later weeks suggests some temporal drift, which a random split alone would hide.